Supplementary material for UAI 2026 Submission 642
DP-SGD analysis: membership inference under differential privacy

This document presents ROC curves for four membership inference attacks (LiRA, BASE, BaVarIA-n, BaVarIA-t) on models trained with DP-SGD at varying noise levels ($\sigma \in \{0, 0.01, 0.1, 1.0\}$, clipping norm $C = 10$).

All experiments use $K = 64$ shadow models, 32 bootstrap replicates, and the online (IN/OUT-aware) setting. Four testbeds are evaluated: CIFAR-10/WRN, CIFAR-100/WRN, Location/MLP3, Purchase-100/MLP3.

Curves show mean ROC ± 1 SE over 32 replicates on log-log axes. Legend entries include AUC.

1 Experimental setup

The models used in this analysis were trained with a separate pipeline calibrated specifically for DP-SGD compatibility, differing from the standard testbeds used in the main paper in several respects:

- **Normalization.** For the WideResNet (WRN) image models, BatchNorm layers are replaced with GroupNorm. DP-SGD requires per-sample gradient computation, which is incompatible with batch-level statistics used by BatchNorm. GroupNorm computes channel-group statistics per sample and is the standard replacement in the DP-SGD literature.
- **Batch size.** Larger batches are used (e.g. 1024 for CIFAR-10/100 vs. smaller batches in the main paper) to improve the signal-to-noise ratio of gradient updates under DP-SGD noise injection. Learning rates also had to be retuned to accommodate.

These changes apply to *all* models in the pipeline, including the undefended baselines (σ=0), so that the only variable across defence levels is the noise multiplier σ. As a consequence, baseline accuracy and risk metrics differ from the main-paper testbeds.

The image testbeds suffer a substantial accuracy drop (-12.7 and -27.5 percentage points for CIFAR-10 and CIFAR-100, respectively), which is attributable primarily to the BatchNorm $\rightarrow$ GroupNorm switch. The tabular testbeds (Location, Purchase), which use plain MLP architectures without any normalization layers, are unaffected by this change and in fact benefit from the retuned hyperparameters (larger batch size and adjusted learning rate), yielding modest accuracy improvements.

Because the baseline models are already weaker for the image testbeds, the absolute membership-inference risk metrics in the following tables and figures are not directly comparable to those in the main paper; they should be interpreted *relative* to each other across noise levels within this document.

2 Privacy parameters and model utility

Table 1 summarises the (ε, δ)-DP guarantees achieved by the Opacus accountant and the resulting test-set accuracy. Delta is set to $\delta \approx 1/n_{\text{train}}$ for each dataset. At σ=0.01 the privacy budget is vacuous ($\varepsilon > 100$K); at σ=1.0 the guarantee becomes meaningful ($\varepsilon \approx 6$–9) but utility degrades substantially, especially for CIFAR-100 and Location.

Table 1: Privacy parameters and model utility for each testbed and noise level. $C = 10$ for all DP-SGD runs. ε rounded down to nearest order of magnitude when > 100.

Testbed	Defence	Test Acc (%)	ε	δ
CIFAR-10 / WRN	Baseline	76.1	—	—
	$\sigma = 0.01$	71.6	>1M	3.3×10^{-5}
	$\sigma = 0.1$	71.2	>10K	3.3×10^{-5}
	$\sigma = 1.0$	54.1	6.7	3.3×10^{-5}
CIFAR-100 / WRN	Baseline	40.0	—	—
	$\sigma = 0.01$	27.7	>1M	3.3×10^{-5}
	$\sigma = 0.1$	26.4	>10K	3.3×10^{-5}
	$\sigma = 1.0$	9.0	6.1	3.3×10^{-5}
Location / MLP3	Baseline	66.4	—	—
	$\sigma = 0.01$	64.6	>100K	4.0×10^{-4}
	$\sigma = 0.1$	55.8	>1K	4.0×10^{-4}
	$\sigma = 1.0$	7.1	8.9	4.0×10^{-4}
Purchase / MLP3	Baseline	89.6	—	—
	$\sigma = 0.01$	87.3	>1M	1.0×10^{-5}
	$\sigma = 0.1$	86.4	>1K	1.0×10^{-5}
	$\sigma = 1.0$	77.3	5.7	1.0×10^{-5}

3 Defence-level view

Each figure shows one defence level (noise setting) with all four attack methods overlaid. Panel titles include mean test accuracy across shadow models.

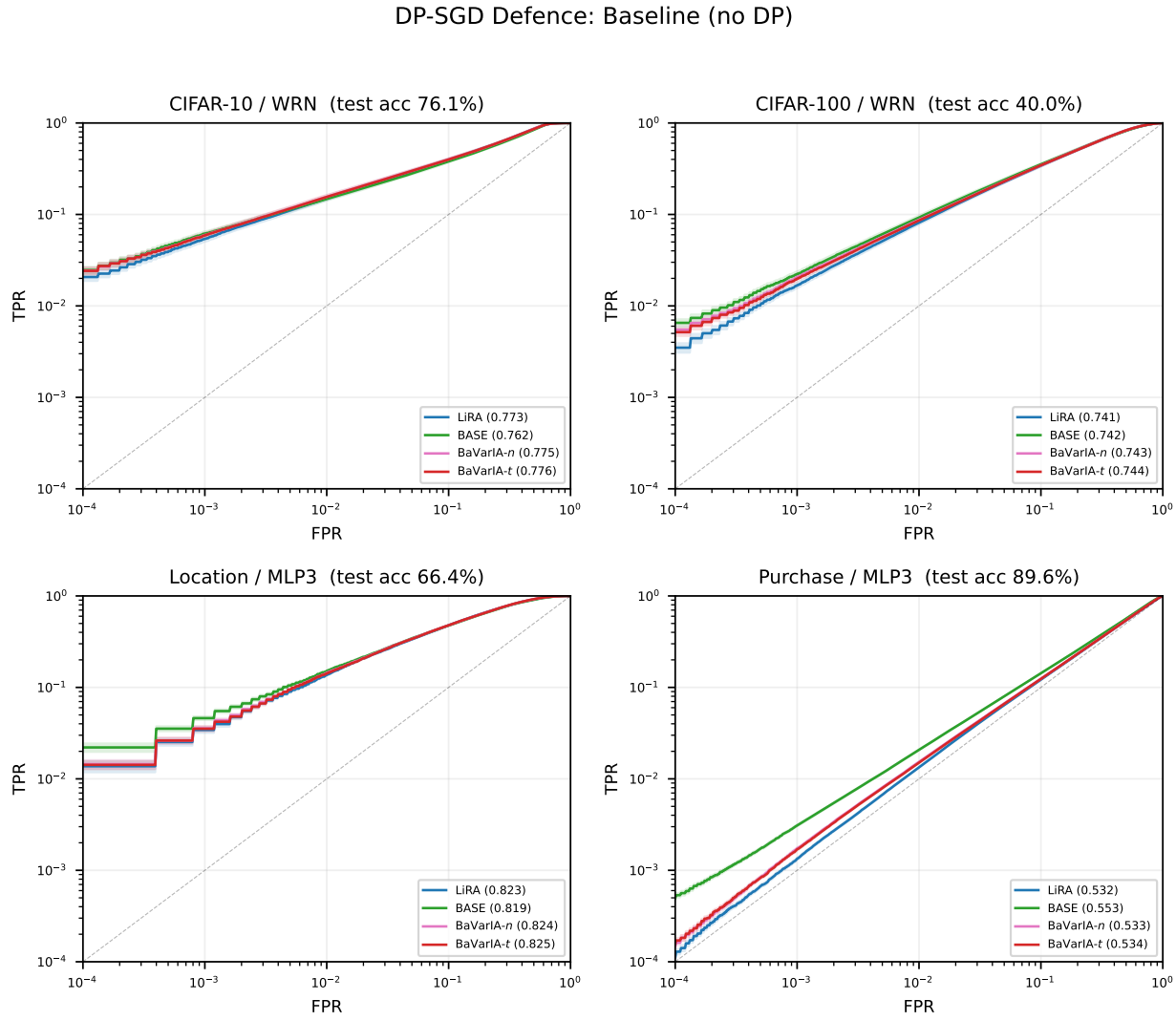

Figure 1: Baseline (no DP-SGD): all four attacks on each testbed.

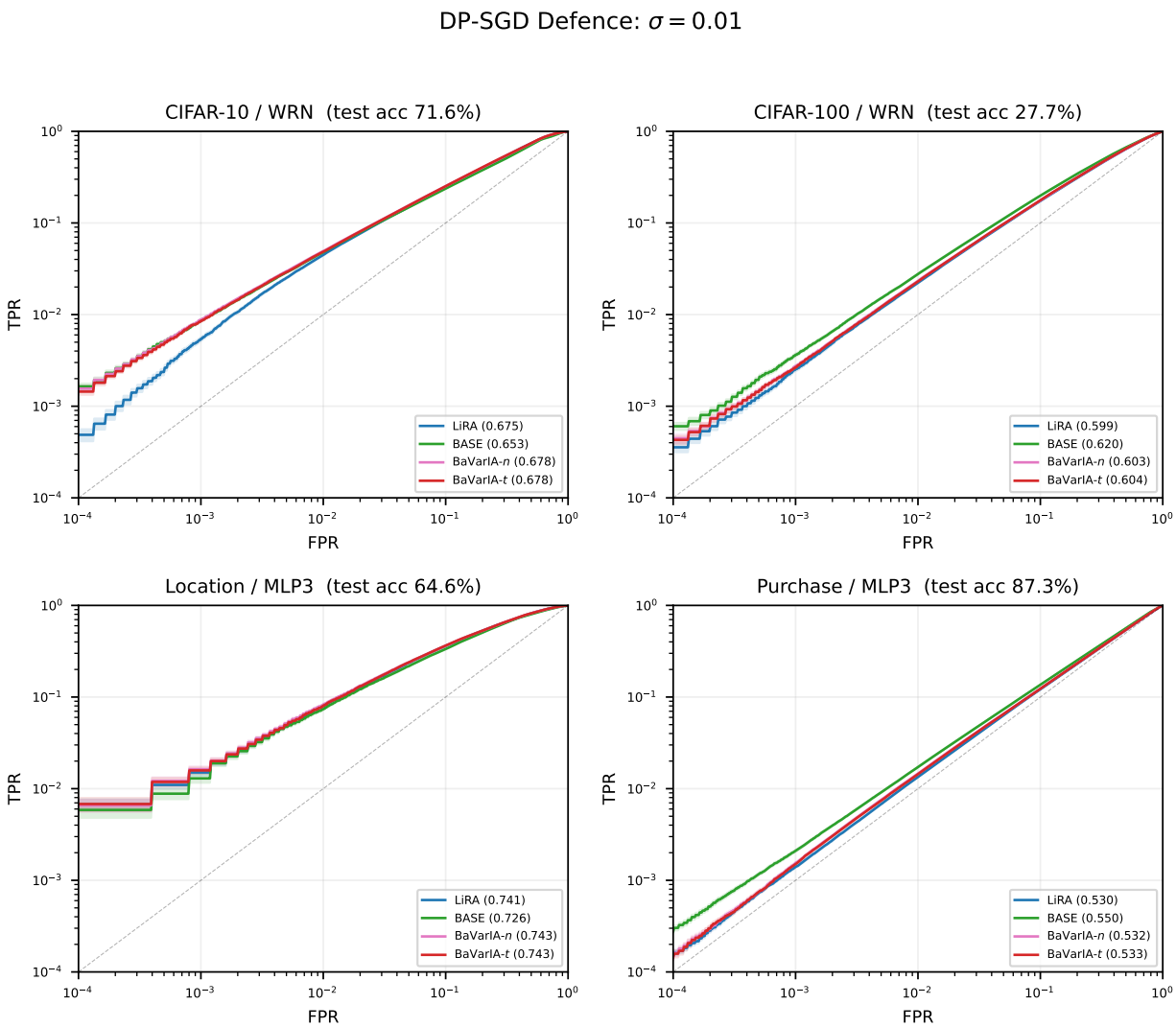

Figure 2: DP-SGD with $\sigma = 0.01$, $C = 10$.

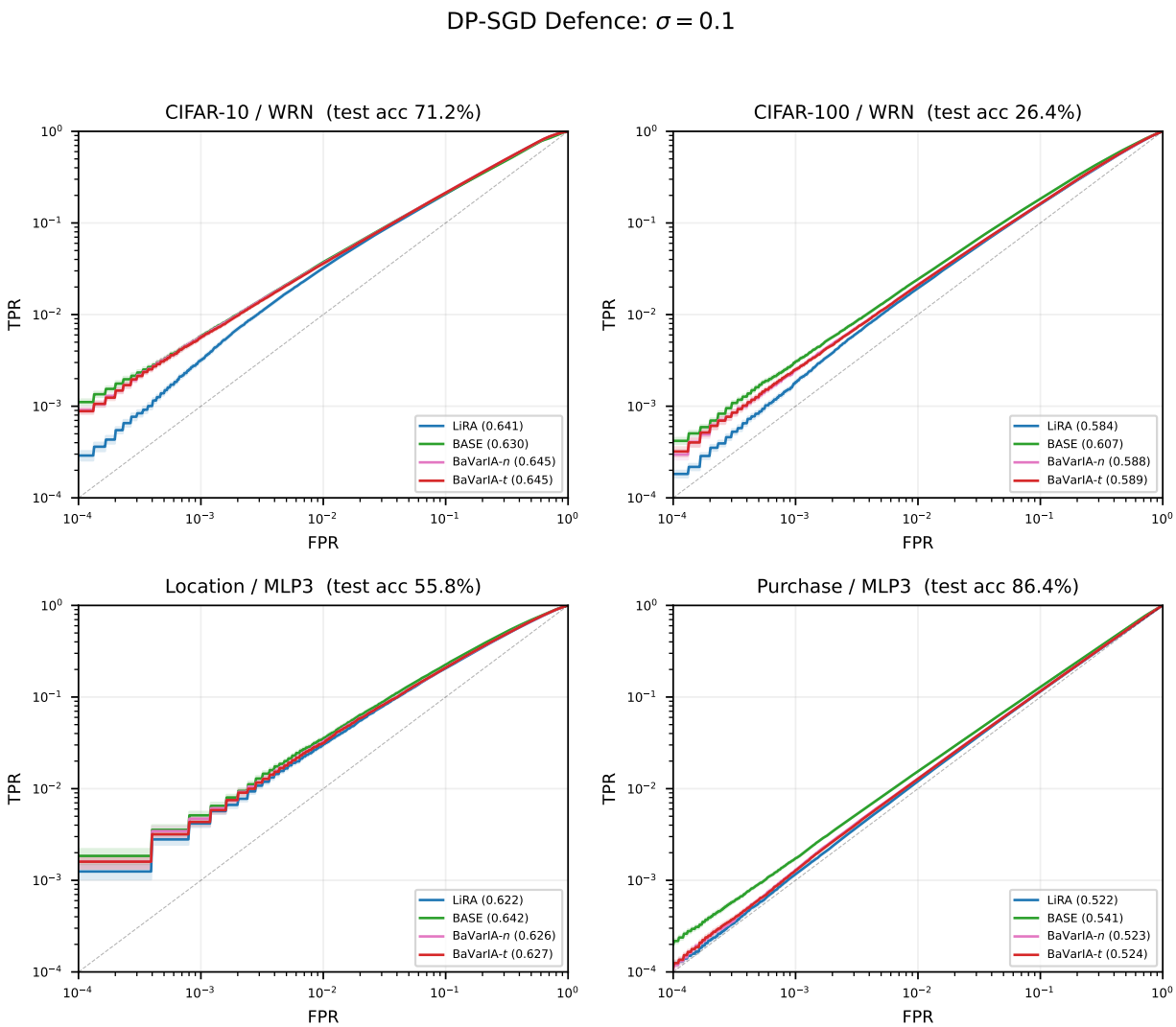

Figure 3: DP-SGD with $\sigma = 0.1$, $C = 10$.

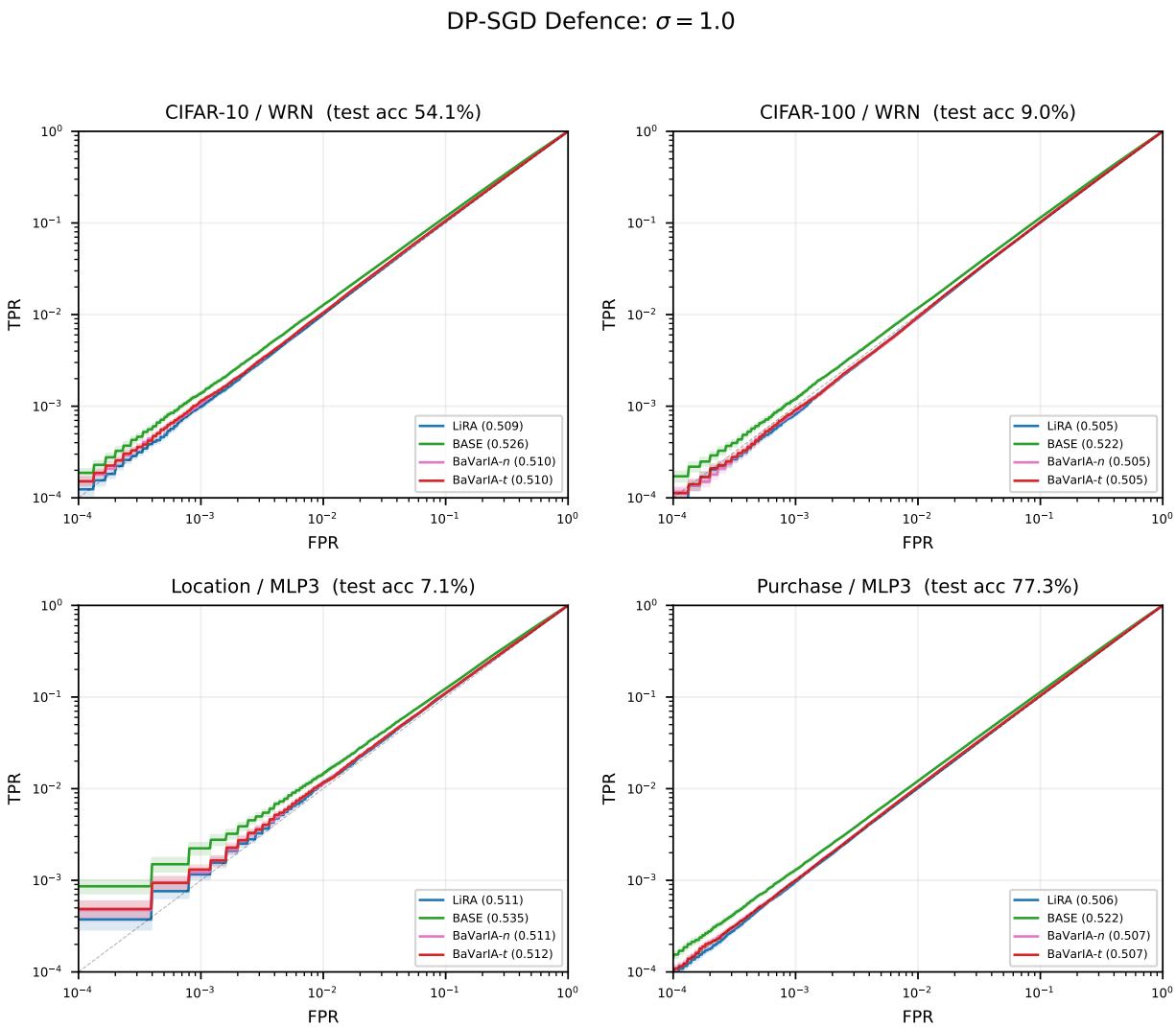

Figure 4: DP-SGD with $\sigma = 1.0$, $C = 10$.

4 Attack-method view

Each figure shows one attack method with all four defence levels overlaid, using that method's canonical colour (LiRA: blue, BASE: green, BaVarIA-n: pink, BaVarIA-t: red) with linestyle distinguishing defence levels.

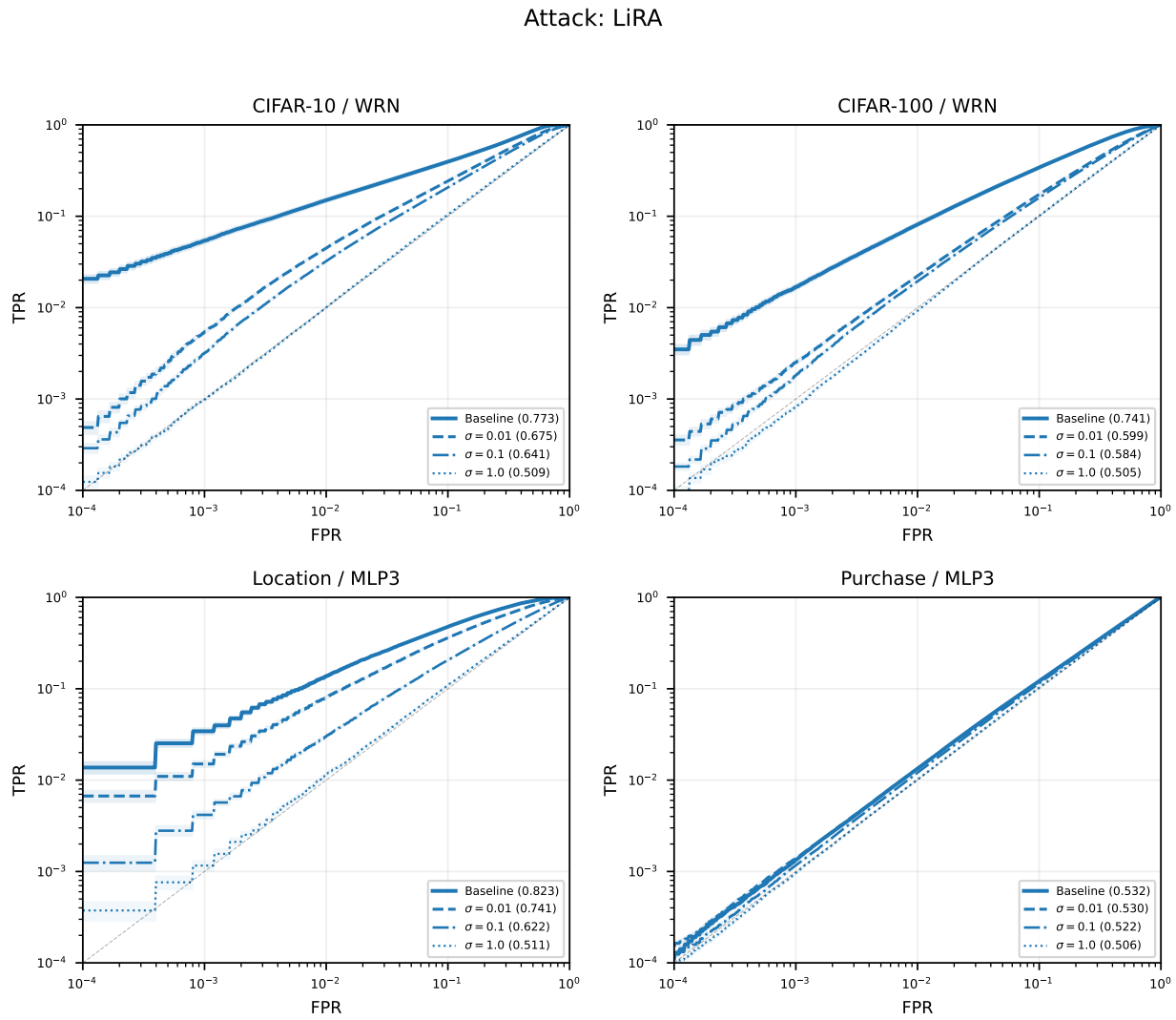

Figure 5: LiRA across defence levels.

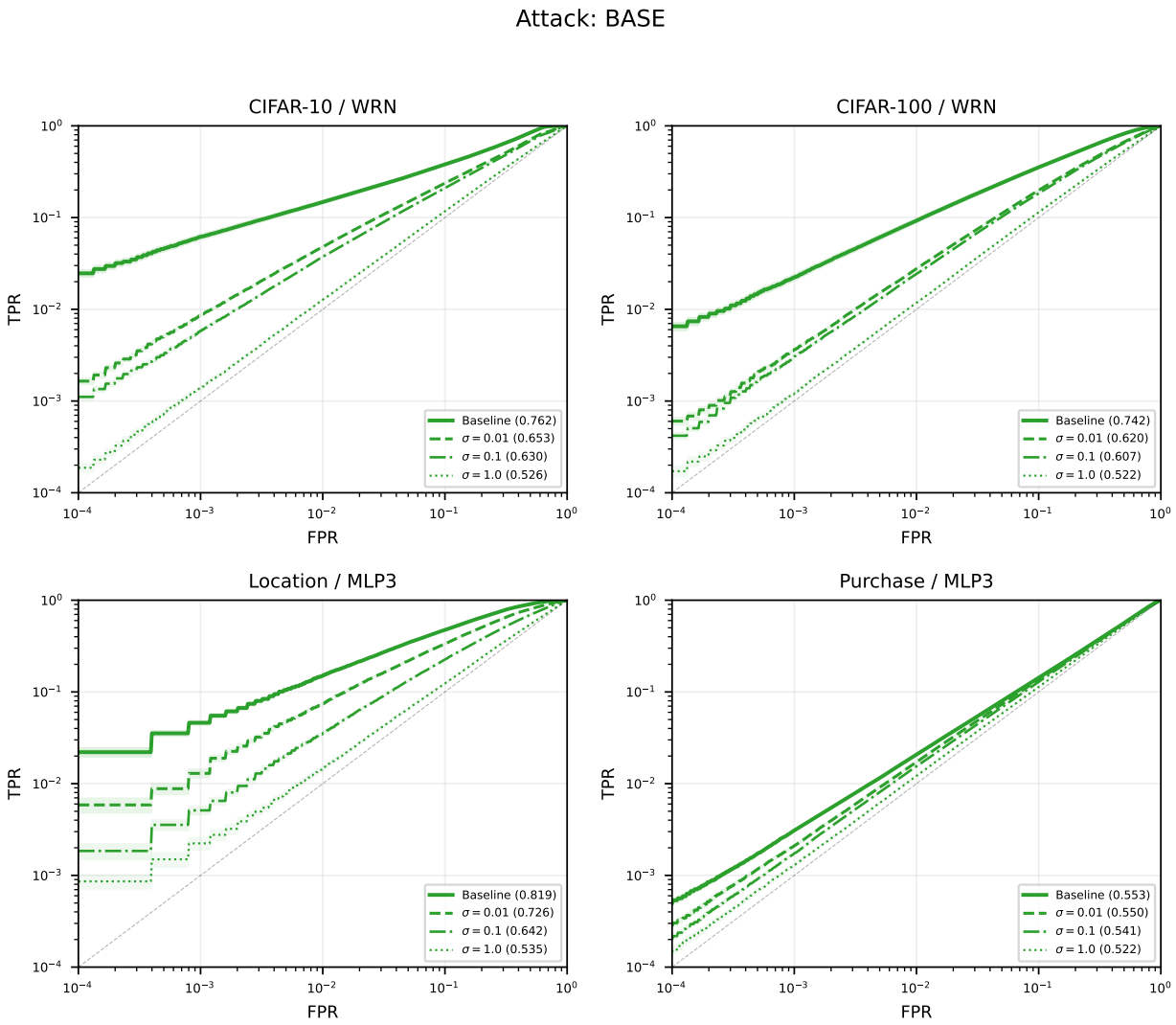

Figure 6: BASE across defence levels.

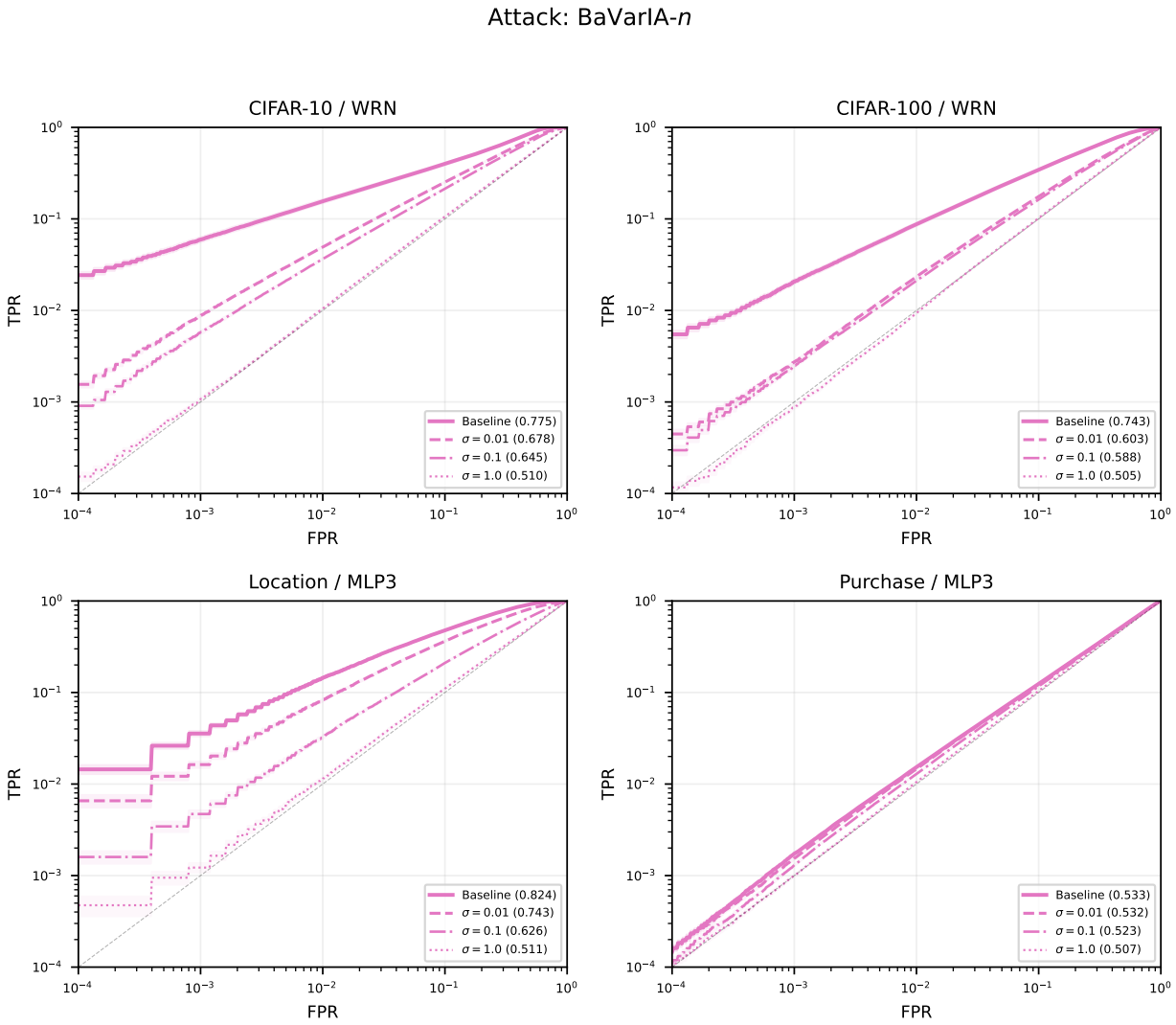

Figure 7: BaVarIA-n across defence levels.

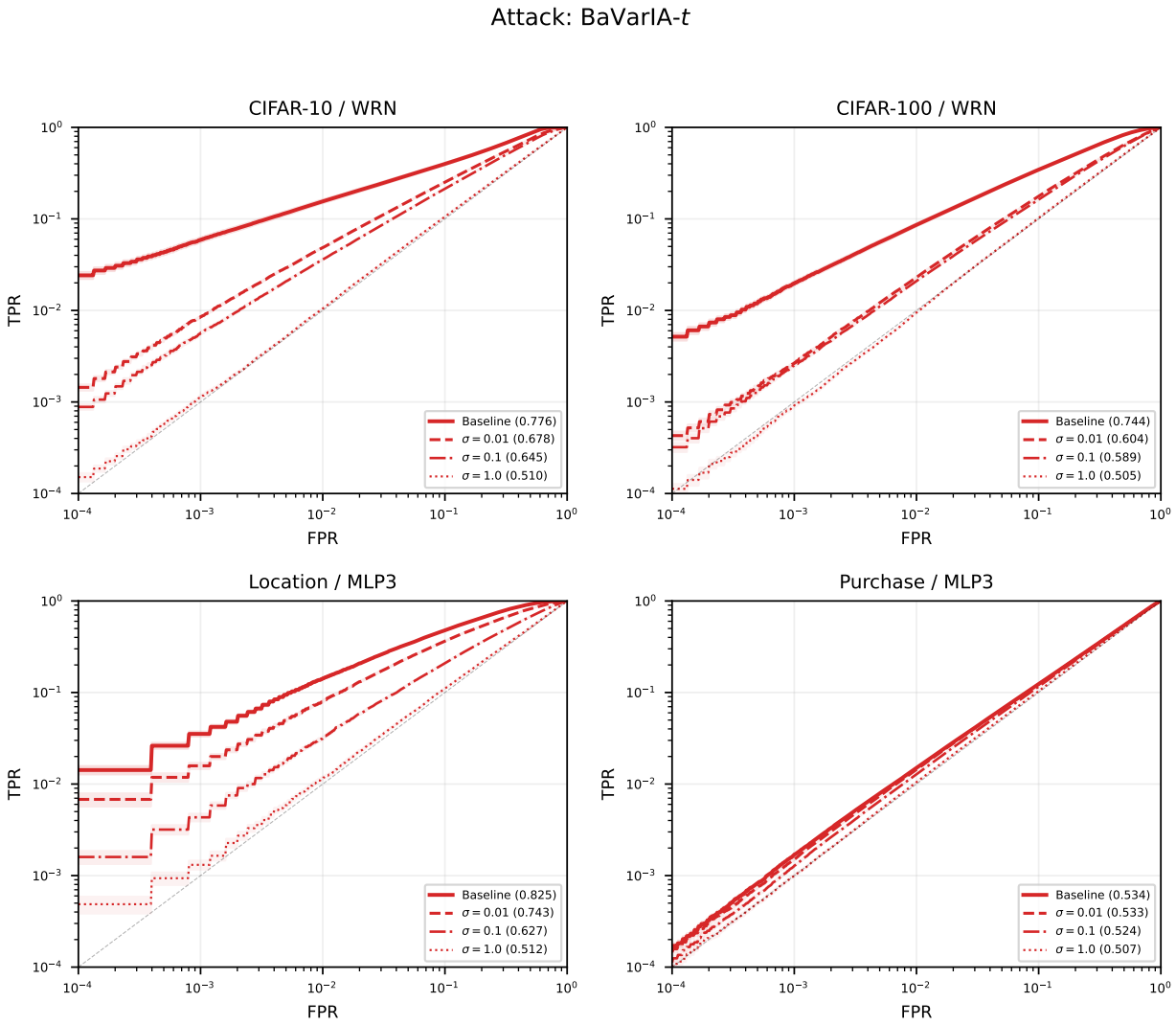

Figure 8: BaVarIA-t across defence levels.

5 Metric tables

Tables 2–4 report means over 32 replicates (K=64, online setting). All digits shown are significant (SE < 0.5 in the last displayed digit). Best method per row in bold.

Table 2: AUC (mean over 32 replicates).

Testbed	Defence	Acc (%)	LiRA	BASE	BaVarIA-n	BaVarIA-t
CIFAR-10 WRN	Baseline	76.1	.773	.762	.775	**.776**
	$\sigma = 0.01$	71.6	.674	.653	**.678**	.678
	$\sigma = 0.1$	71.2	.641	.630	**.645**	.645
	$\sigma = 1.0$	54.1	.509	**.526**	.510	.510
CIFAR-100 WRN	Baseline	40.0	.741	.742	.743	**.744**
	$\sigma = 0.01$	27.7	.600	**.620**	.603	.604
	$\sigma = 0.1$	26.4	.584	**.607**	.588	.589
	$\sigma = 1.0$	9.0	.505	**.522**	.505	.505
Location MLP3	Baseline	66.4	.823	.819	.824	**.825**
	$\sigma = 0.01$	64.6	.741	.726	.743	**.743**
	$\sigma = 0.1$	55.8	.622	**.642**	.626	.627
	$\sigma = 1.0$	7.1	.511	**.535**	.511	.512
Purchase MLP3	Baseline	89.6	.532	**.553**	.533	.534
	$\sigma = 0.01$	87.3	.530	**.550**	.532	.533
	$\sigma = 0.1$	86.4	.522	**.541**	.523	.524
	$\sigma = 1.0$	77.3	.506	**.522**	.507	.507

Table 3: TPR at FPR=0.01 (mean over 32 replicates).

Testbed	Defence	Acc (%)	LiRA	BASE	BaVarIA-n	BaVarIA-t
CIFAR-10 WRN	Baseline	76.1	.150	.148	**.156**	.156
	$\sigma = 0.01$	71.6	.045	.048	**.049**	.049
	$\sigma = 0.1$	71.2	.032	**.038**	.037	.036
	$\sigma = 1.0$	54.1	.010	**.013**	.011	.011
CIFAR-100 WRN	Baseline	40.0	.082	**.093**	.087	.086
	$\sigma = 0.01$	27.7	.022	**.028**	.023	.023
	$\sigma = 0.1$	26.4	.019	**.024**	.021	.021
	$\sigma = 1.0$	9.0	.009	**.012**	.009	.010
Location MLP3	Baseline	66.4	.139	**.153**	.146	.144
	$\sigma = 0.01$	64.6	.081	.075	**.084**	.081
	$\sigma = 0.1$	55.8	.031	**.036**	.033	.032
	$\sigma = 1.0$	7.1	.012	**.015**	.012	.012
Purchase MLP3	Baseline	89.6	.013	**.021**	.015	.015
	$\sigma = 0.01$	87.3	.013	**.017**	.015	.014
	$\sigma = 0.1$	86.4	.012	**.016**	.013	.013
	$\sigma = 1.0$	77.3	.010	**.012**	.011	.011

Table 4: TPR at FPR=0.001 (mean over 32 replicates).

Testbed	Defence	Acc (%)	LiRA	BASE	BaVarIA-n	BaVarIA-t
CIFAR-10 WRN	Baseline	76.1	.054	**.062**	.060	.059
	$\sigma = 0.01$	71.6	.005	.009	**.009**	.009
	$\sigma = 0.1$	71.2	.003	**.006**	.006	.006
	$\sigma = 1.0$	54.1	.0010	**.0014**	.0011	.0012
CIFAR-100 WRN	Baseline	40.0	.017	**.022**	.021	.020
	$\sigma = 0.01$	27.7	.003	**.004**	.003	.003
	$\sigma = 0.1$	26.4	.002	**.003**	.002	.003
	$\sigma = 1.0$	9.0	.0008	**.0012**	.0009	.0009
Location MLP3	Baseline	66.4	.034	**.046**	.036	.035
	$\sigma = 0.01$	64.6	.015	.013	**.016**	.016
	$\sigma = 0.1$	55.8	.004	**.005**	.005	.004
	$\sigma = 1.0$	7.1	.0012	**.0022**	.0012	.0013
Purchase MLP3	Baseline	89.6	.0013	**.0031**	.0017	.0017
	$\sigma = 0.01$	87.3	.0014	**.0021**	.0015	.0015
	$\sigma = 0.1$	86.4	.0012	**.0017**	.0013	.0013
	$\sigma = 1.0$	77.3	.0010	**.0013**	.0010	.0010

6 Observations

BaVarIA consistently dominates LiRA. Both BaVarIA-n and BaVarIA-t outperform LiRA in AUC on all four testbeds at every noise level tested, including the strongest defence (σ=1.0). The margin is largest at baseline (up to +0.003 AUC) and shrinks under heavy noise but never reverses.

BASE leads under DP-SGD noise. As the noise multiplier increases, BASE progressively overtakes the Bayesian methods. At baseline BaVarIA wins 3 of 4 testbeds; at σ=0.01 the race is even (2–2); at σ=0.1 BASE leads on 3 of 4; and at σ=1.0 BASE wins all four. The likely explanation is that BASE uses the simpler exponential (single-parameter) model on losses, which is more robust when the signal-to-noise ratio collapses. The Gaussian models underlying LiRA and BaVarIA must estimate per-sample means and variances that become unreliable under heavy gradient noise. The absolute differences remain small throughout ($\leq$ 0.025 AUC, $\leq$ 0.005 TPR@0.01).

Hierarchy-ordering principle. The crossover — BaVarIA-n holding its advantage over LiRA at every noise level while BASE / RMIA overtakes both Gaussian methods as σ grows — mirrors the ordering observed at small shadow-model budgets in the absence of any defence: simpler distributional families dominate whenever the per-sample signal is scarce, whether the scarcity comes from few shadow models or from added DP-SGD noise. The pattern is therefore not specific to DP training but a general property of the parameter-sharing hierarchy linking BASE / RMIA, BaVarIA, and LiRA.

All attacks approach random guessing at σ=1.0. At the strongest defence level, AUC margins above chance (0.5) range from +0.005 (LiRA on CIFAR-100) to +0.035 (BASE on Location). TPR at FPR=0.01 is $\leq$0.015 for every method–testbed pair, compared to a random baseline of 0.01. This confirms that DP-SGD with σ=1.0 ($\varepsilon \approx$ 6–9) effectively neutralises membership inference, though it comes at a steep utility cost (CIFAR-100 drops to 9% accuracy, Location to 7%).

Privacy–utility–leakage trade-off. The intermediate regime σ=0.01 is notable: it degrades model accuracy only modestly (CIFAR-10: 76%→72%, Purchase: 90%→87%) while substantially reducing attack AUC (e.g. CIFAR-10: 0.776 → 0.678). However, the formal privacy guarantee at this noise level is vacuous ($\varepsilon >$ 1M). The practical privacy may stem from gradient clipping (C=10) rather than noise injection.